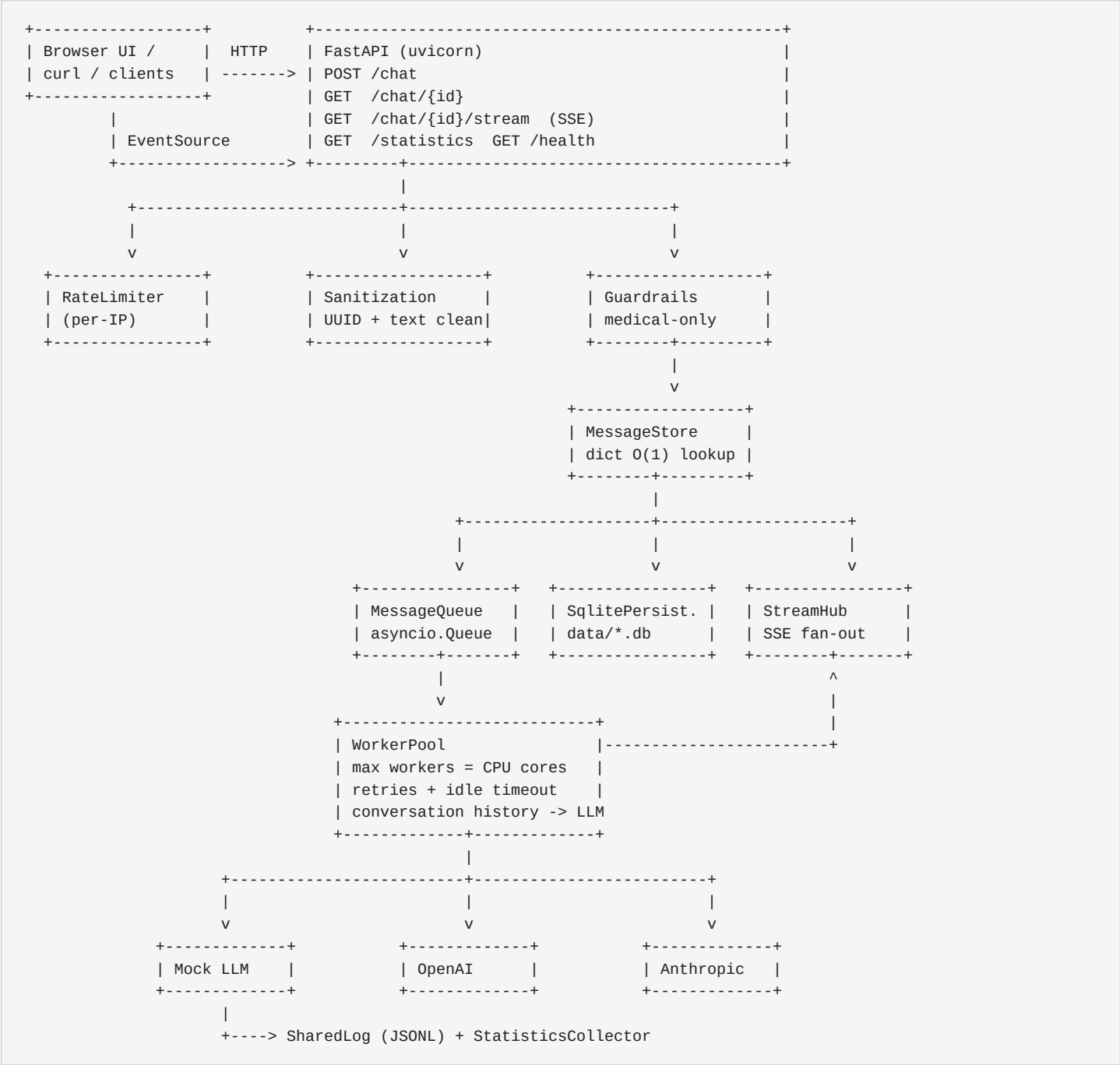


Medical Expert AI Chat

Current system architecture

1. High-level architecture

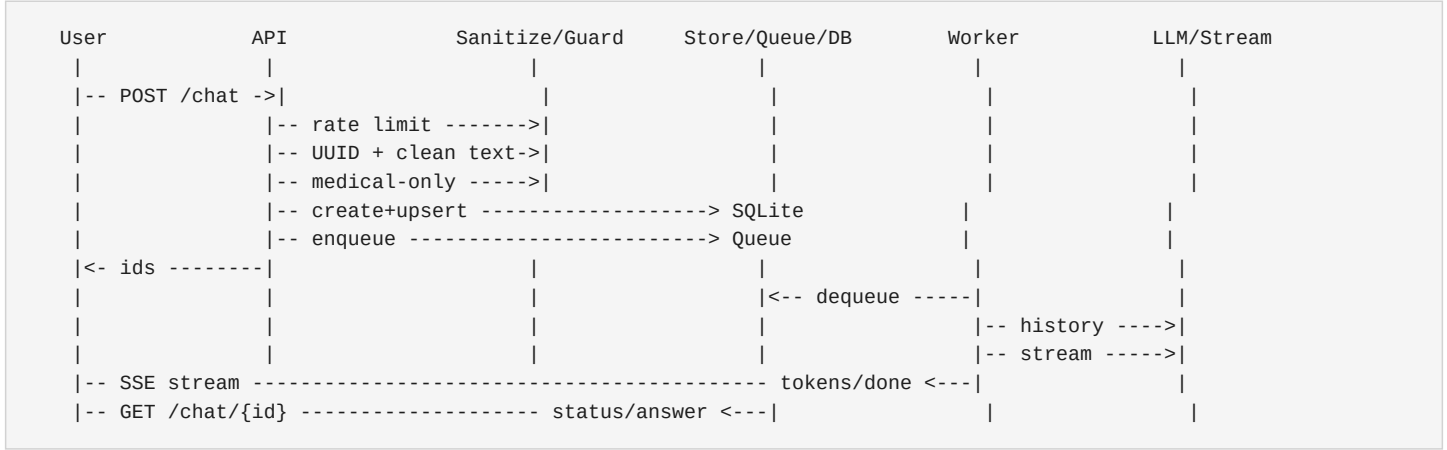
Python 3.13 FastAPI service with async worker pool, SQLite persistence, conversation history (agent-mode), SSE streaming, medical guardrails, input sanitization, rate limiting, shared file logs, and statistics.



2. Request lifecycle

POST /chat validates and sanitizes input, optionally continues a conversation, persists the message, enqueues work, and returns messageId + conversationId. Clients poll GET /chat/{id} or stream GET /chat/{id}/stream.

Medical Expert AI Chat - Architecture



3. Module map

Item	Description
main.py	App factory, lifespan, static UI
api.py	HTTP routes + SSE stream
config.py	Env Settings (pydantic-settings)
sanitization.py	UUID checks, text/null-byte cleaning
guardrails.py	Reject non-medical / abusive input
rate_limit.py	Per-IP sliding-window limiter
storage.py	In-memory store + queue + file log
persistence.py	SQLite upsert/load across restarts
stream_hub.py	Fan-out token/done/error events
worker_pool.py	Workers, retries, history, idle exit
statistics.py	O(1) metrics counters
llm/	mock / OpenAI / Anthropic clients
static/	UI with agent-mode + EventSource

4. Design choices

Item	Description
Async processing	API returns immediately; workers run in background
Agent-mode	Optional conversationId; prior Q&A sent to LLM
Persistence	SQLite (bound params); MessageStore dict for O(1)
Streaming	SSE token/done/error via StreamHub
Safety	Sanitize + UUID validate; no string-built SQL
Concurrency	asyncio queue + tasks; max = CPU cores
Logs	Append-only JSON lines file (thread-safe)
Config	Environment variables via Settings
Deploy	venv + medical-chat / uvicorn; GET /health
Python	Requires Python 3.13+

5. Data status lifecycle

```
pending --> processing --> completed
                \--> failed
```

Each transition updates MessageStore + SQLite.
Completed/failed also append SharedLog, update Statistics, and emit StreamHub done/error.

6. HTTP API

Item	Description
POST /chat	Submit question (+ optional conversationId)
GET /chat/{id}	processing completed+answer failed+error
GET /chat/{id}/stream	SSE: token / done / error
GET /statistics	Queue, workers, retries, tokens, timings
GET /health	Health check
GET /	Simple chat UI

7. Safety notes

User input is sanitized (NFKC, control/null stripping, length bounds). conversationId and messageId must be UUIDs. SQLite uses only bound parameters. Conversation history reused in prompts is re-sanitized and capped.